Auto Ticket Classification using Flow Designer

Solution Requirements

1. PURPOSE

This document lists what the solution must do and how well it must do it. The requirements come from the empathy map, the problem definition, the customer journey and the data flow diagram, and they guide the design and build phases.

2. SOLUTION OVERVIEW

The solution is an event-driven workflow built in Flow Designer. When a new ticket is created, the flow reads the description, matches keywords against configurable rules, sets Category, Subcategory, Urgency and Priority, assigns the correct resolver group, and attaches the SLA timer. Tickets that match no rule go to a review queue for Tier-1 agents.

3. FUNCTIONAL REQUIREMENTS

ID	Functional Requirement	Priority	Traced to
FR1	The flow must trigger automatically when a new Incident or Request item is created.	High	DFD 1.0; Empathy map
FR2	The system must scan the short description and detailed body text for keywords such as VPN, Password Reset and Hardware Failure.	High	DFD 2.0
FR3	Identified terms must be matched against configurable keyword and decision rules.	High	DFD 3.0; Root cause
FR4	The system must set Category and Subcategory automatically from the matched rule.	High	Problem definition
FR5	The system must set Urgency and Priority from the rule, using the same value for identical issues.	High	Inconsistent prioritization
FR6	The ticket must be assigned to the correct resolver group (Network, Security, Identity Management, Database).	High	DFD 4.0; Ping-pong issue
FR7	The assigned group must be notified and the correct SLA timer attached on dispatch.	High	DFD 5.0; SLA goal
FR8	A ticket with no rule match must go to a default review queue for a Tier-1 Agent and never stay unassigned.	High	Customer journey
FR9	Every automated decision (matched rule, category, group) must be recorded on the ticket as a work note.	Medium	Audit and trust
FR10	The IT Manager must be able to view routing accuracy, re-assignment rate and SLA compliance.	Medium	Success metrics

4. NON-FUNCTIONAL REQUIREMENTS

ID	Category	Requirement	Measure
NFR1	Performance	Classification and routing must complete right after ticket creation.	Under 10 seconds
NFR2	Accuracy	First-time assignment to the correct group must be reliable.	90% or higher
NFR3	Maintainability	Admins must update keywords and rules without writing code.	Low-code, no deployment
NFR4	Scalability	The flow must handle peak load without queue build-up.	100+ tickets per day
NFR5	Reliability	If the flow fails, the ticket must still land in the review queue.	Zero lost tickets
NFR6	Security	Only authorised roles may edit flows and rule tables.	Role-based access
NFR7	Auditability	All flow executions must be logged and reviewable.	100% logged

5. TECHNICAL AND PLATFORM REQUIREMENTS

Component	Requirement
Platform	Enterprise ITSM platform with Flow Designer enabled and a development instance for testing.
Trigger	Record-created trigger on the Incident table and the Request Item table.
Flow actions	Look up records, update record fields, assign to group, send notification and attach SLA.
Rule storage	A keyword and decision rule table holding keyword, category, subcategory, priority and group.
Reference data	Existing assignment group and SLA definition tables read by the flow.
Test data	A standard set of sample incidents covering VPN, password, hardware, database and security issues.

6. SAMPLE CLASSIFICATION RULE MATRIX

These sample rules show how keywords map to fields and groups. The final list will be configured and tuned during the build phase.

Keyword	Category	Subcategory	Priority	Assignment Group
VPN, remote access	Network	VPN	High	Network Operations
Password Reset, locked out	Access	Password	Moderate	Identity Management
Hardware Failure, not booting	Hardware	Laptop / Desktop	Moderate	Desktop Support
Database down, SQL error	Database	Availability	Critical	Database Administration
Phishing, malware	Security	Incident	High	Security Operations

7. USER ROLES AND ACCESS

Role	Access needed
Flow Administrator	Create and edit the flow and the keyword and decision rule table.
Service Desk (Tier-1) Agent	View classified tickets and work the default review queue.
Resolver Group Member	Receive routed tickets and alerts, and request re-assignment if needed.
IT Manager	View dashboards and reports; approve rule and SLA policy changes.

8. ASSUMPTIONS AND CONSTRAINTS

- **Assumption:** the platform already has Flow Designer, assignment groups and SLA definitions available.
- **Assumption:** users describe the issue in short text, so keyword matching is enough for the first release.
- **Constraint:** the solution works only on standard ticket fields and does not need a non-standard legacy system.
- **Constraint:** hardware repair, field dispatch and manual fixes for custom code bugs are out of scope.
- **Constraint:** the first release uses rule-based logic, not machine learning.

9. ACCEPTANCE CRITERIA

Criterion	Test	Pass condition
Auto trigger	Create 20 sample incidents and requests.	Flow runs on all 20
Classification	Run the standard incident dataset.	90% or more correct
Routing	Check the assignment group on each ticket.	Re-routing below 5%
Fallback	Submit tickets with vague text ("System error").	All reach review queue
Speed	Measure time from creation to assignment.	Under 10 seconds

10. CONCLUSION

These requirements cover the full path from ticket creation to routing and notification. Meeting them removes manual triage, raises routing accuracy above 90%, and reduces SLA breaches, which solves the main problem found in the empathy map and problem definition.